SATYARTH MISHRA

AI/ML Engineer | Computer Vision | Edge AI

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PROFESSIONAL SUMMARY

Computer Science & AI student at GLA University specializing in Computer Vision, Edge AI, and ML application development. Built and deployed YOLOv8-based real-time detection systems on Raspberry Pi and ESP32-CAM, achieving approximately 88% detection accuracy and 15+ FPS on edge hardware. Experienced with Python, PyTorch, OpenCV, FastAPI, AWS, embedded systems, and data-driven ML pipelines. Also developing AI product and interface design skills through applied projects.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, Java
AI/ML: PyTorch, TensorFlow, YOLOv8, OpenCV, scikit-learn, Keras, MobileNetV3, Deep Learning, Machine Learning
Computer Vision: Object Detection, Image Classification, Image Preprocessing, Real-Time Inference, Model Evaluation
Data: NumPy, Pandas, Matplotlib, Data Preprocessing, Exploratory Data Analysis (EDA), SQL
Edge & Embedded: Raspberry Pi, ESP32, ESP32-CAM, Arduino, Edge AI Inference, Model Quantization, Model Optimization
Robotics & Hardware: DC Motors, Servo Motors, L298N Motor Drivers, Ultrasonic Sensor, MQ2 Gas Sensor, Flame Sensor, DHT Sensor
Backend & APIs: FastAPI, REST API, MongoDB, MySQL
Cloud & DevOps: AWS EC2, AWS S3, Git, GitHub, GitHub Actions, Linux, Vercel, Netlify
Protocols: Wi-Fi, UART, I2C, SPI, PWM
Additional: Qiskit, Quantum Circuits, Quantum Gates, Basic Quantum Algorithms

EXPERIENCE

AI/ML & Embedded Systems Developer — Jan 2026 – Present

VEENA Smart Rover — Raspberry Pi, ESP32, Python, OpenCV, YOLOv8, NumPy

- Engineered an autonomous disaster rescue rover integrating ultrasonic, MQ2, flame, and temperature sensing with real-time AI decision making.
- Deployed a YOLOv8 person-detection pipeline on ESP32-CAM, achieving approximately 88% detection accuracy under simulated disaster lighting conditions.
- Developed and validated the Computer Vision pipeline on Raspberry Pi using Python, OpenCV, NumPy, and Pandas, then migrated the system to ESP32 and ESP32-CAM.
- Achieved 15+ FPS real-time inference and reduced hardware cost by approximately 70% while maintaining real-time performance.

SELECTED PROJECTS

VEENA — Autonomous Disaster Rescue Rover — Jan 2026 – Present

Raspberry Pi, ESP32, ESP32-CAM, Python, OpenCV, YOLOv8, NumPy, C++

- Designed and built a low-cost autonomous rescue rover for earthquake, flood, and fire scenarios targeting confined, debris-filled environments.
- Integrated multi-sensor fusion across four sensor types with explainable AI logic for real-time hazard classification, achieving sub-200ms response latency from sensor trigger to actuator output.
- Deployed YOLOv8-based person detection on ESP32-CAM with approximately 88% accuracy for victim identification under low-visibility conditions and Wi-Fi alerts to an operator dashboard.
- Used NumPy and Pandas for sensor preprocessing, time-series analysis of environmental readings, and model performance evaluation.
- Architected a modular hardware system with four DC motors, two servo motors, two L298N motor drivers, and a water pump for mobility and fire suppression.
- Migrated the complete AI/CV pipeline from Raspberry Pi to ESP32/ESP32-CAM, achieving 70% lower hardware cost at equivalent 15+ FPS real-time inference.
- Implemented two-way communication using ESP32 Wi-Fi, LCD, and speaker for operator alerts and victim interaction in zero-visibility environments.

Repository: github.com/foxDev-code/veena

Smart Farming Assistant — Edge AI — 2026

MobileNetV3, YOLOv8n, Python, Computer Vision, Edge AI

- Built a 27-class crop-disease dataset covering major crops including maize, wheat, cotton, rice, and sugarcane.
- Dataset split: 12,414 training images, 1,554 validation images, and 1,562 test images.
- Trained MobileNetV3-Large for crop-disease classification and YOLOv8n for pest detection under edge-AI constraints.
- Designed an offline-first product concept for low-connectivity environments incorporating weather intelligence, crop risk, irrigation, farmer alerts, multilingual assistance, mandi information, and government-scheme analysis.
- Explored AI product architecture connecting model predictions to practical farmer-facing workflows and decision support.

AI-Powered Web Application with ML Backend — Mar 2026 – Apr 2026

Python, FastAPI, MongoDB, AWS EC2, NumPy, Pandas, JavaScript

- Architected a full-stack AI web application with a FastAPI REST API serving ML model predictions and deployed it on AWS EC2.
- Built ML data-processing workflows using NumPy and Pandas for preprocessing, feature engineering, and model-ready datasets.
- Designed MongoDB schemas for inference logs and user sessions with indexed query performance for concurrent access.
- Integrated a pretrained NLP model through an API, reducing manual input processing time by approximately 50%.
- Configured GitHub Actions CI for automated testing and deployment to AWS on pushes to the main branch.

Repository: github.com/foxDev-code/ai-webapp

Quantum Computing Explorations — 2026

Qiskit, Python

- Implemented quantum circuits using Hadamard, CNOT, and Pauli gates and demonstrated superposition and entanglement.
- Explored Deutsch-Jozsa and Grover’s search algorithm as part of self-directed quantum computing study.

PRODUCT & DESIGN — ADDITIONAL CAPABILITY

- AI Product Design: Designed feature architecture, workflows, dashboards, alerts, chatbot interactions, and decision-support concepts for AI-powered applications.
- UI/UX: Worked on interface concepts, information hierarchy, cards, user flows, and dashboard experiences for web and AI products.
- Visual Design: Created posters, certificates, workshop creatives, social-media graphics, QR registration assets, and event communication for technical/student initiatives.

- Creative Technology: Combined AI/ML engineering with product and visual design; photography and visual storytelling maintained as a supporting creative practice.

EDUCATION

GLA University — Bachelor of Technology, Computer Science and Artificial Intelligence Aug 2025 – Jul 2029
Noida, UP, India

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900)
- Microsoft Fabric: DP-600 / DP-700 learning and exam preparation

COMMUNITY

- Synapse AI Club: Contributed to AI-focused workshops and student technical initiatives, including event design, certificates, posters, and promotional creatives.